

Reihaneh Malekian

Ph.D. Candidate in Statistics, Columbia University
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Education

Columbia University, New York, NY

Expected May 2027

Ph.D. in Statistics

Advisor: Sumit Mukherjee

Sharif University of Technology, Tehran, Iran

M.Sc. in Pure Mathematics

Sharif University of Technology, Tehran, Iran

B.Sc. in Mathematics and Applications

Research Interests

Probability and high-dimensional statistics; large deviations; random graphs and hypergraphs; Gibbs measures and interacting systems; statistical robustness and inference for dependent and structured data; empirical Bayes.

Publications and Preprints

- **LDP for Tensor Forms**, with Sohom Bhattacharya, Nabarun Deb, and Sumit Mukherjee. 2026
arXiv:2609.02682.
- **Is Speckle Noise More Challenging to Mitigate Than Additive Noise?**, with Hao Xing and Arian Maleki. *IEEE Transactions on Information Theory*, 71(8):6353–6372. 2025
- **Matrix Concentration: Order versus Anti-order**, with Aaditya Ramdas. *arXiv:2408.05998*. 2024

Manuscripts in Preparation

- “Fluctuations in Mean-Field Potts Models,” with Sumit Mukherjee.
- “Statistical Inference for Online Least-Squares SGD under Stationary Dependence,” with Promit Ghosal.
- “Empirical Bayes Inference with Gaussian Scale Mixture Priors,” with Taehyun Kim.

Selected Research Presentations

- Invited talk, MIT Probability Seminar, Massachusetts Institute of Technology, November 2026.
- Talk, Yale Statistics Student Seminar, Yale University, October 2026.
- Invited talk, Probability Seminar, UC Berkeley, September 2026.
- Contributed talk, Joint Statistical Meetings, Boston, August 2026.
- Talk, ICERM Hot Topics Workshop, Brown University, August 2026.
- Poster, International Congress of Mathematicians, Philadelphia, July 2026.
- Poster, Peter Hall and Bob Shumway Memorial Conference on Dependent Data Analysis in the Modern Era, UC Davis, May 2026.
- Invited talk, Finger Lakes Probability Seminar, Syracuse University, April 2026.
- Seminar talk, Columbia University Student Seminar, February 2026.
- Talk, Progress in Discrete and Continuous Probability Conference, Union College, Schenectady, NY, July 2025.
- Poster, INFORMS Applied Probability Society Conference, Atlanta, July 2025.
- Poster, Bernoulli–IMS World Congress in Probability and Statistics, Bochum, Germany, August 2024.
- Talk, Minghui Yu Memorial Conference, Columbia University, April 2024, April 2025 and April 2026.

Honors and Awards

- Teaching Observation Fellow, Columbia University Center for Teaching and Learning, 2026–2027.
- Early Career Travel Award, American Statistical Association, 2026.
- Graduate Research Fellowship, Columbia University, 2022–2027.
- Ranked second in overall GPA in both the B.Sc. and M.Sc. cohorts, Sharif University of Technology.
- Recipient of undergraduate and graduate scholarships from Iran’s National Elites Foundation.
- Admitted to the graduate program without an entrance examination as an exceptionally talented B.Sc. student, Sharif University of Technology.
- Placed among the top 0.05% of participants in Iran’s nationwide university entrance examination.

Selected Workshops and Summer Schools

ICERM Hot Topics Workshop Synergies between Geometry, Probability, and Computation in High Dimensions, Brown University	August 2026
SLMath Summer Graduate School Concentration Inequalities and Localization Techniques in High-Dimensional Probability and Geometry	July 2023

Teaching Experience

Teaching Assistant , Columbia University	2022–Present
• Introduction to Statistics; Statistical Inference; Linear Regression Models; Introduction to Statistical Reasoning; Introduction to Probability and Statistics; Probability.	
Teaching Assistant , Sharif University of Technology	
• Probability and Applications; Probability Theory; Mathematical Writing; Algebra I; Analysis I.	

Research and Academic Mentoring

Directed Reading Program Mentor , Columbia University	Fall 2026
Mentoring an undergraduate student in probability through an individualized semester-long reading project.	
Bachelor’s Project Co-supervisor , EPFL	2021
Co-supervised Clémence Marcaillou’s bachelor’s project in combinatorial discrepancy theory through weekly research meetings.	

Selected Expository Writing

- “Good Numbers.” *Pargar*, Year 2, no. 4, pp. 17–23, Spring 2014. [In Persian]
- “The Turán Number and Probabilistic Combinatorics,” by Alan J. Aw. Persian translation and adaptation of the article published in *The American Mathematical Monthly*, 119(6):510–513, 2012. *Pargar*, Year 2, no. 8, pp. 9–13, Spring 2015. [In Persian]

Outreach and Academic Service

Graduate Student Mentor , GROW 2026, Columbia University	2026
First-Year PhD Peer Mentor , Department of Statistics, Columbia University	2023, 2024, 2026
Organizing Team Member and Program Moderator , Maryam Mirzakhani Celebration, Sharif University of Technology, Tehran, Iran	May 2019
Student Organizer , Undergraduate Mathematics Summer School, IASBS, Zanjan, Iran	2019
Student Organizer , Mathematics Summer School for High School Students, Borujen, Iran	2015

Skills

Programming: Python, R.
Languages: Persian (Farsi; native), English (proficient), French (intermediate, B1).